No significant linear trend in pre-landfall typhoon speed and direction along Southeast China coast (1980–2024): a multi-scale decomposition perspective

Yongyan Ni
11359683

Supervisor: David Schultz

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The University of Manchester

ABSTRACT

在全球变暖背景下，热带气旋的移动速度和移动方向受到广泛关注，但其**区域尺度、特别是在登陆前高暴露窗口内的变化趋势仍不明确**。本研究聚焦 1980—2024 年间登陆中国东南沿海的台风，分析其在登陆前 24 小时内的移动速度和主导方向的变化。基于 CMA 最佳路径数据对 336 个台风事件进行**简单线性回归**，未检测到移动速度（斜率 -0.006 km/h/年，p = 0.830）和西北方向占比（斜率 0.362%/年，p = 0.068）存在显著的长期趋势。

然而，采用**集合经验模态分解（EEMD）**（Huang et al., 1998；Wu and Huang, 2009）进行多尺度分析后，进一步揭示了显著的**年际（类 ENSO，2–7 年）** 和**准年代际振荡（QDO，9–18 年）** 信号，两种台风特征的长期趋势分量均表现出**复杂的非线性多年代际波动**。两个序列中均未识别出显著的太平洋年代际振荡（PDO，20–30 年）分量。

这些结果表明，**全球尺度的线性减速信号（Kossin, 2018）在近岸登陆前窗口中并未简单重现**。相反，区域动力过程（包括复杂近岸地形、海陆热力差异和西北太平洋副热带高压形态）对信号产生了显著的**调制（modulate）** 作用，可能掩盖了微弱的线性趋势。因此，在中国东南沿海台风灾害风险评估中，不能直接套用全球线性趋势，而必须系统考虑多尺度气候振荡的影响。

Against the backdrop of global warming, the movement speed and direction of tropical cyclones have received widespread attention, but their regional-scale trends, particularly within the high exposure window before landfall, remain unclear. This study focuses on typhoons that made landfall along the southeastern coast of China between 1980 and 2024, analyzing changes in their movement speed and dominant direction in the 24 hours prior to landfall. Simple linear regression analysis of 336 typhoon events based on CMA optimal track data revealed no significant long-term trends in movement speed (slope -0.006 km/h/year, p = 0.830) or the proportion of northwesterly directional cyclones (slope 0.362%/year, p = 0.068).

However, multi-scale analysis using ensemble empirical mode decomposition (EEMD) (Huang et al., 1998; Wu and Huang, 2009) further revealed significant interannual (ENSO-like, 2–7 years) and quasi-decadal oscillations (QDO, 9–18 years) signals. The long-term trend components of both typhoon characteristics exhibited complex nonlinear multi-decadal fluctuations. No significant Pacific decadal oscillation (PDO, 20–30 years) component was identified in either series.

These results indicate that the global-scale linear deceleration signal (Kossin, 2018) did not simply reappear in the pre-landfall window near the coast. Instead, regional dynamic processes (including complex nearshore topography, land-sea thermal differences, and the Northwest Pacific subtropical high morphology) significantly modulated the signal, potentially masking the weak linear trend. Therefore, in typhoon disaster risk assessment along the southeast coast of China, global linear trends cannot be directly applied; the impact of multi-scale climate oscillations must be systematically considered.

Contribution statement (Required)

The optimal typhoon track data used in this study were obtained with the assistance of my supervisor (or the name of the data provider). All data processing, statistical analysis, coding, and visualization were completed independently by myself. I wrote the first draft of the paper, and my supervisor David Schultz provided guidance on research design, interpretation of results, and manuscript revision.

AI usage statement (Required)

During this research, I used the following artificial intelligence tools to assist in completing some non-core tasks: 1. Google Gemini: Used to create a research timeline and break down tasks into a list of actionable steps. 2. Microsoft Copilot: Used to assist in writing, modifying, and debugging Python code, providing programming suggestions, and guiding the upload of data and code to GitHub. 3. Deepseek: Used to assist in carefully reading references (summarizing core content, explaining technical concepts, and providing feedback and corrections to my own understanding), as well as polishing the paper text and checking formatting and terminology consistency.

1. Introduction

在全球变暖背景下，热带气旋（Tropical Cyclone, TC）的活动特征正在发生深刻变化。其中，移动速度（translation speed）直接影响区域过境时长及灾害累积（Kossin, 2018）。Kossin（2018）发现 1949–2016 年西北太平洋 TC 平均移速显著下降约 10%，并将其归因于人为变暖导致的引导气流（steering flow）减弱。

Against the backdrop of global warming, the activity characteristics of tropical cyclones (TCs) are undergoing profound changes. Among these changes, translation speed directly affects the duration of their passage through the region and the accumulation of hazards (Kossin, 2018). Kossin (2018) found that the average translation speed of TCs in the Northwest Pacific decreased significantly by about 10% from 1949 to 2016, attributing this to a weakening of steering flow caused by anthropogenic warming.

然而，全球尺度的统计趋势能否直接外推到区域尺度，尤其是登陆前的近岸窗口，尚不明确。近岸区域受地形摩擦、海陆热力对比及副热带高压边缘强引导气流影响，其动力机制与开阔洋面存在差异。此外，现有研究多使用全球数据集（如 IBTrACS）或分析全生命周期，缺乏针对中国东南沿海“登陆前 24 小时”这一高暴露窗口的系统趋势检测。这是一个知识空白（knowledge gap），因为在这一高暴露窗口内台风移动的精确特征对区域防灾准备和风险评估至关重要（Liu et al., 2022）。虽然中国气象局（CMA）最佳路径数据在近岸区域具有最高的定位精度（Ying et al., 2014），但这一窗口内的多尺度变率尚未被系统研究。

However, it remains unclear whether global-scale statistical trends can be directly extrapolated to regional scales, especially the nearshore window before landfall. Nearshore regions are influenced by topographic friction, land-sea thermal contrast, and strong steering airflows at the edge of the subtropical high, resulting in dynamic mechanisms that differ from those of the open ocean. Furthermore, existing studies primarily utilize global datasets (such as IBTrACS) or analyze the entire typhoon lifecycle, lacking systematic trend detection for the high-exposure window of the 24 hours before landfall along China's southeast coast. This represents a knowledge gap, as the precise characteristics of typhoon movement within this high-exposure window are crucial for regional disaster preparedness and risk assessment (Liu et al., 2022). Although the China Meteorological Administration (CMA) optimal path data has the highest positioning accuracy in the nearshore region (Ying et al., 2014), the multiscale variability within this window has not been systematically studied.

本研究使用 CMA 最佳路径数据（1980–2024 年）。选择该时段主要基于三点考虑：第一，1980 年后静止卫星实现常规覆盖，数据均一性和可靠性显著提高，避免了早期观测手段变化带来的非均一性问题（Ying et al., 2014）；第二，45 年时段覆盖了完整的太平洋年代际振荡（PDO）位相转换及多次强 ENSO 事件，长度足以支撑后续的多尺度分解，同时避免了卫星时代之前可能存在的台风漏记问题；第三，数据更新至 2024 年，包含了近年极端台风事件，对当前灾害风险评估具有参考意义。据我所知，尚无前人研究使用 EEMD 在此特定窗口内检验台风移速和方向的多尺度变率。因此，为了超越简单线性趋势分析的局限性，并完整捕捉这一关键高暴露窗口内的复杂多尺度变率，本研究采用**集合经验模态分解（EEMD）**。

This study uses CMA optimal path data (1980–2024). This timeframe was chosen primarily based on three considerations: First, geostationary satellites achieved regular coverage after 1980, significantly improving

data uniformity and reliability, thus avoiding the non-uniformity issues caused by changes in early observation methods (Ying et al., 2014); second, the 45-year period covers the complete Pacific Decadal Oscillation (PDO) phase transition and multiple strong ENSO events, providing sufficient length to support subsequent multi-scale decomposition while avoiding potential typhoon omissions from the pre-satellite era; third, the data is updated to 2024, including recent extreme typhoon events, providing reference for current disaster risk assessment.To the best of my knowledge, no previous studies have used EEMD to examine the multi-scale variability of typhoon speed and direction within this specific window.Therefore, in order to overcome the limitations of simple linear trend analysis and fully capture the complex multiscale variability within this critical high-exposure window, this study employs ensemble empirical mode decomposition (EEMD).

本研究分析登陆中国东南沿海的台风在登陆前 24 小时内的移动速度与方向变化趋势，提出两个工作假设：

This study analyzes the trends in the translation speed and direction of typhoons making landfall along the southeastern coast of China in the 24 hours prior to landfall, and proposes two working hypotheses:

H1a：移动速度显著降低；

H1b：主导移动方向发生显著变化（如西北向比例增加或向极地偏移）。

H1a: translation speed is significantly reduced;
H1b: The dominant movement direction changes significantly (e.g., the proportion of northwestward movement increases or the movement shifts towards the poles).

本研究的独特贡献在于，通过使用 **EEMD 多尺度分析**超越简单线性趋势检测，**从而直接回应了在复杂区域背景下线性方法的局限性**。它系统揭示了 1980—2024 年中国东南沿海登陆前 24 小时窗口内台风移动速度和西北方向占比的复杂年际和年代际变率结构。这强调了在评估区域台风灾害风险时，必须充分考虑这些多尺度自然变率的影响，而不仅仅依赖于全球尺度或全生命周期内的单一线性趋势。

The unique contribution of this study lies in its use of EEMD multiscale analysis to move beyond simple linear trend detection, thus directly addressing the limitations of linear methods in complex regional contexts. It systematically reveals the complex interannual and interdecadal variability structures of typhoon movement speed and the proportion of northwestward movement within the 24-hour window preceding landfall along China's southeast coast from 1980 to 2024. This emphasizes that when assessing regional typhoon disaster risk, the impact of these multiscale natural variability must be fully considered, rather than relying solely on a single linear trend at the global scale or throughout the entire typhoon lifecycle.

需要说明的是，本研究仅采用简单的线性回归与频数统计，方法并不复杂。但这正是科学进步的一种基本模式：每一位研究者贡献一个透明、可重复的简单分析，并通过开放数据与代码共享信息，无数这样的微小工作汇集起来，就能使人类逐步理解全球变暖对热带气旋影响这一极其复杂的系统。本研究的全部代码与中间数据均已公开，以方便后续研究在此基础上继续推进。

It should be noted that this study only employed simple linear regression and frequency statistics, and the methods were not complex. However, this is precisely a fundamental model of scientific progress: each researcher contributes a transparent, reproducible, and simple analysis, and information is shared through open data and code. Countless such small efforts, accumulated, enable humanity to gradually understand the extremely complex system of the impact of global warming on tropical cyclones. All the code and intermediate data from this study have been made public to facilitate further research based on this foundation.

2. Data, Materials, and Methods

本章节描述了分析 1980 至 2024 年间登陆中国东南沿海地区台风移动特征的研究方法。主要包括数据来源、处理流程、登陆事件的定义、分析的数据来源的台风的空间和时间范围，移动 velocity（speed 与 direction）的计算方式，以及所使用的统计分析方法。所有分析均基于可复现的 Python 脚本完成。

This chapter describes the research methodology for analyzing the movement characteristics of typhoons that made landfall in southeastern coastal China between 1980 and 2024. It mainly includes data sources, processing procedures, the definition of landfall events, the spatial and temporal range of the typhoons in the data sources, the calculation methods for movement velocity (speed and direction), and the statistical analysis methods used. All analyses were performed using reproducible Python scripts.

2.1 Data Source

本研究使用的数据是台风最佳路径数据（Best-track data），来源是中国气象局（China Meteorological Administration, CMA）。数据时间范围为 1980 年至 2024 年，该数据覆盖了西太平洋盆地内所有生成并影响中国的热带气旋。原始 CMA 数据为结构不统一的文本格式（.txt），每个文件对应一个年份的台风轨迹记录。数据集中包含台风编号（SID）、国际名称（NAME）、观测时间（ISO_TIME）、台风类型代码（TYPE_CODE）、经纬度（LAT/LON）、中心气压（PRESSURE）和最大风速（WIND_MS）等变量。(Lu et al., 2020; Ying et al., 2014)

The data used in this study is best-track typhoon data, sourced from the China Meteorological Administration (CMA). The data spans from 1980 to 2024, covering all tropical cyclones that formed in the western Pacific basin and affected China. The original CMA data was in a non-uniform text format (.txt), with each file corresponding to a typhoon track record for one year. The dataset includes variables such as typhoon number (SID), international name (NAME), observation time (ISO_TIME), typhoon type code (TYPE_CODE), latitude and longitude (LAT/LON), central pressure (PRESSURE), and maximum wind speed (WIND_MS). (Lu et al., 2020; Ying et al., 2014)

2.2 Data Pre-processing and Filtering

首先使用了自定义的 Python 脚本解析原始 CMA 数据。解析逻辑包括识别每个台风的头部信息（包含台风 ID 和名称）及其对应的轨迹点数据行。解析后，数据被转换为结构化的 Pandas DataFrame 格式，并新增年份（YEAR）和以(km/h)为单位的风速（WIND_KMH）字段。

First, a custom Python script was used to parse the raw CMA data. The parsing logic included identifying the head information of each typhoon (including typhoon ID and name) and its corresponding trajectory point data rows. After parsing, the data was converted into a structured Pandas DataFrame format, and new fields for year and wind speed in km/h (WIND_KMH) were added.

为聚焦于影响中国的台风事件，数据集被进行筛选，筛选纬度范围为 15.0°N 至 40.0°N，经度范围为 105.0°E 至 130.0°E。筛选后，共识别出 681 个在该区域内活动的台风事件。

To focus on typhoon events affecting China, the dataset was filtered, with latitude ranging from 15.0°N to 40.0°N and longitude ranging from 105.0°E to 130.0°E. After filtering, a total of 681 typhoon events active in this region were identified.

2.3 Definition of Landfall and Analysis Window

由于缺少精确的海岸线矢量数据，本研究采用地理边界框（Geospatial Bounding Box）方法近似识别台风在中国的登陆事件。该方法通过在目标区域（如中国东南沿海）定义一个矩形经纬度范围，将首次进入该范围且满足强度条件的台风轨迹点判定为登陆点。尽管该方法在精度上低于使用 GIS Shapefile 计算海岸线交点的方法，但其实现简便、计算高效，适用于大规模、长时间的初步研究。

Due to the lack of precise coastline vector data, this study employs the geospatial bounding box method to approximate typhoon landfall events in China. This method defines a rectangular latitude and longitude range within a target area (such as the southeastern coast of China) and identifies the landfall point as the first point in the trajectory of a typhoon that enters this range and meets the intensity criteria. Although this method is less accurate than the method using GIS Shapefiles to calculate coastline intersections, it is simple to implement, computationally efficient, and suitable for large-scale, long-term preliminary studies.

边界框与强度条件设置如下：

The bounding box and strength conditions are set as follows:

- 纬度范围：18.0°N – 35.0°N
- 经度范围：108.0°E – 123.0°E
- 强度阈值：台风在进入边界框时，其强度至少达到热带风暴（Tropical Storm, TS）级别，即风速 ≥ 17.5 (m/s)（或 63 (km/h)），对应 CMA 数据中的类型代码（TYPE_CODE）≥ 1。
- ▪ Latitude range: 18.0°N – 35.0°N
- ▪ Longitude range: 108.0°E – 123.0°E
- Intensity threshold: When a typhoon enters the boundary box, its intensity must reach at least the level of a tropical storm (TS), i.e., wind speed ≥ 17.5 (m/s) (or 63 (km/h)), corresponding to a type code (TYPE_CODE) ≥ 1 in the CMA data.

对于每个台风，沿其轨迹顺序扫描，将第一个同时满足上述地理与强度条件的轨迹点定义为登陆点（Landfall Point）。所有后续分析（包括移动速度和方向）均基于每个台风的登陆点及其前 24 小时（pre-landfall 24-hour）内的轨迹数据。在此定义下，共识别出 336 个符合条件的台风登陆事件。

For each typhoon, its trajectory was scanned sequentially, and the first trajectory point that simultaneously met the above geographical and intensity conditions was defined as the landfall point. All subsequent analyses (including translation speed and direction) were based on each typhoon's landfall point and its trajectory data for the preceding 24 hours. Under this definition, a total of 336 typhoon landfall events meeting the criteria were identified.

2.4 Calculation of Typhoon Movement Characteristics

2.4.1 translation speed

台风移动速度（Translation Speed）通过计算两个连续轨迹点之间的距离与时间差获得。距离计算采用 Haversine 公式，该公式适用于球面几何，可准确估算地球表面两点间的大圆距离。公式如下：

The translation speed of a typhoon is obtained by calculating the distance and time difference between two consecutive points on its trajectory. The distance is calculated using the Haversine formula, which is applicable to spherical geometry and can accurately estimate the great circle distance between two points on the Earth's surface. The formula is as follows:

$$a = \sin(\frac{\Delta\phi}{2}) + \cos(\phi_1)\cdot\cos(\phi_2)\cdot\sin^2(\frac{\Delta\lambda}{2})$$
$$c = 2\cdot\mathrm{atan2}(\sqrt{a}, \sqrt{1-a})$$
$$d = R\cdot c$$

其中:

- a 为中间变量，表示半正矢值(haversine)的平方和，无量纲;
- ϕ 为纬度（latitude）, 单位为弧度（radians）;
- λ 为经度（longitude），单位为弧度（radians）;
- c 为中心角(central angle), 即两点与地心连线之间的夹角，单位为弧度(radians);
- R 为地球平均半径，取 6371 (km);
- d 为两点之间的距离，单位为公里（km）。

Where:

a is an intermediate variable, representing the sum of squares of the haversine values, dimensionless;

ϕ is latitude, in radians;

λ is longitude, in radians;

c is the central angle, the angle between the line connecting the two points to the Earth's center, in radians;

R is the Earth's average radius, taken as 6371 (km);

d is the distance between the two points, in kilometers (km).

时间差（单位：小时）由相邻轨迹点的 ISO_TIME 字段计算得出。速度计算公式为：

The time difference (in hours) is calculated from the ISO_TIME field of adjacent trajectory points. The speed calculation formula is:

$$\boxed{\text{speed}(km \cdot h^{-1}) = \frac{\text{distance(km)}}{\text{time_difference}(h)}}$$

每个台风的登陆前 24 小时平均移动速度定义为该时段内所有有效分段速度的算术平均值。
The average translation speed of each typhoon in the 24 hours before landfall is defined as the arithmetic mean of all effective segment speeds during that period.

关于后续图表中时间序列表示的说明

在展示台风特征（如平均移动速度、西北方向占比）随时间变化的图表中，**每个数据点代表该年内所有相关台风事件的年度聚合值。**例如，图 6 中的“原始速度(Original Speed)”黑线显示的是年度平均移动速度，由多个独立台风事件计算得出。这些年度值的连线（如图 4 和图 6 中的黑线）呈现出自然的**年际变率**，由于年度聚合的离散性和台风事件本身的变异性，线条可能显得不连续或呈“锯齿状”。这些视觉特征准确反映了作为多尺度分解输入的年度数据。

Explanation of Time Series Representation in Subsequent Charts:
In charts showing how typhoon characteristics (such as average speed and northwestward proportion) change over time, each data point represents the annual aggregate value of all relevant typhoon events within that year. For example, the black line "Original Speed" in Figure 6 shows the annual average speed, calculated from multiple independent typhoon events. The lines connecting these annual values (the black lines in Figures 4 and 6) exhibit natural interannual variability; due to the discreteness of annual aggregates and the variability of the typhoon events themselves, the lines may appear discontinuous or jagged. These visual features accurately reflect the annual data as input for multi-scale decomposition.

2.4.2 Movement Direction

台风移动方向通过计算两个连续轨迹点之间的**方位角（Bearing）**确定。方位角表示从一点到另一点的方向角（以正北为 0°，顺时针增加）。计算公式如下：

The direction of a typhoon's movement is determined by calculating the **azimuth** between two consecutive points on its trajectory. The azimuth represents the direction angle from one point to another (with true north as 0°, increasing clockwise). The calculation formula is as follows:

$$
\begin{aligned}
y &= \sin(\Delta\lambda)\cdot\cos(\phi_2) \\
x &= \cos(\phi_1)\cdot\sin(\phi_2) - \sin(\phi_1)\cdot\cos(\phi_2)\cdot\cos(\Delta\lambda) \\
\theta_{rad} &= \operatorname{atan2}(y, x) \\
\text{bearing} &= (\theta_{\text{deg}} + 360) \mod 360
\end{aligned}
$$

其中：

- y 为方向矢量的北向(或 y 轴)分量，无量纲;
- x 为方向矢量的东向(或 x 轴)分量，单位为弧度（radians）；
- θ_{rad} 为中间变量，从正东逆时针计量的方向角，单位为弧度（radians）；
- θ_{deg} 为单位转换为角度后的 θ_{rad}，单位为角度(degrees);
- bearing 为以正北为 0°，顺时针计量的方位角，单位为角度(degrees)；

Where:

y is the north (or y-axis) component of the direction vector, dimensionless;

x is the east (or x-axis) component of the direction vector, in radians;

θ_{rad} is an intermediate variable, the direction angle measured counterclockwise from due east, in radians;

θ_{deg} is θ rad after converting units to degrees, in degrees;

bearing is the azimuth angle measured clockwise from due north (0°), in degrees.

计算得到的连续方位角随后被分类为 8 个基本方向：北（N）、东北（NE）、东（E）、东南（SE）、南（S）、西南（SW）、西（W）、西北（NW）。
The calculated continuous azimuth angles were then classified into eight basic directions: North (N), Northeast (NE), East (E), Southeast (SE), South (S), Southwest (SW), West (W), and Northwest (NW).

主导移动方向（Dominant Movement Direction）的定义与计算方式如下：在台风登陆前 24 小时窗口内（共包含 4 个 6 小时段），分别计算每个 6 小时段起点至终点的净位移方向（方位角），并将其划分为 8 个基本方向（N, NE, E, SE, S, SW, W, NW）。统计该台风在 4 个时段中出现的方向类别，将出现频次最高的类别（众数）定义为其主导移动方向。若两个类别频次相同，则取第一个出现的最高频次方向。对所有台风重复上述过程，最终得到主导移动方向的频次分布。
(之所以采用 6 小时间隔，是因为 CMA 最佳路径数据集的标准采样频率即为 6 小时（部分早期年份可能为 12 小时，但本研究仅保留连续 6 小时间隔的轨迹点，对于缺失或间隔大于 6 小时的情况，该台风被排除分析)

The definition and calculation method of the Dominant Movement Direction are as follows: Within the 24-hour window before the typhoon's landfall (comprising four 6-hour segments), the net displacement direction (azimuth) from the start to the end of each 6-hour segment is calculated and divided into eight basic directions (N, NE, E, SE, S, SW, W, NW). The directional categories appearing in the four time periods are statistically analyzed, and the category with the highest frequency (mode) is defined as the Dominant Movement Direction. If two categories have the same frequency, the first highest-frequency direction is taken. This process is repeated for all typhoons to obtain the frequency distribution of the Dominant Movement Direction.(The 6-hour interval is used because the standard sampling frequency of the CMA optimal path dataset is 6 hours (it may have been 12 hours in some earlier years, but this study only retains trajectory points with consecutive 6-hour intervals; typhoons with missing intervals or intervals greater than 6 hours are excluded from analysis).

2.5 Statistical Analysis

2.5.1 Trend Analysis of translation speed

为检验台风登陆前平均移动速度是否存在随年份的长期变化趋势，本研究采用简单线性回归（Simple Linear Regression）模型。自变量（independent variable）为年份（Year），因变量（dependent variable）为每个台风的登陆前 24 小时平均移动速度（Avg_Speed_24h_kmh）。回归结果包括斜率（slope）、决定系数（R^2）和 p 值（p-value）。显著性水平设定为 ($\alpha = 0.05$)，即当 ($p < 0.05$) 时认为趋势具有统计显著性。

To examine whether there is a long-term trend in the average speed of typhoons before landfall over the years, this study employed a simple linear regression model. The independent variable was the year, and the dependent variable was the average speed (Avg_Speed_24h_kmh) of each typhoon in the 24 hours before landfall. The regression results included the slope, coefficient of determination (R^2), and p-value. The significance level was set at $\alpha = 0.05$, meaning that a trend was considered statistically significant when $p < 0.05$.

2.5.2 Distribution Analysis of Movement Direction

台风移动方向的分布特征通过频次分布分析（Frequency Distribution Analysis）进行描述。统计每种主导移动方向（N, NE, E, SE, S, SW, W, NW）在 336 个台风事件中出现的次数和百分比。结果通过条形图（Bar Chart / Count Plot）进行可视化。

The distribution characteristics of typhoon movement directions were described using frequency distribution analysis. The frequency and percentage of each dominant movement direction (N, NE, E, SE, S, SW, W, NW) were counted in 336 typhoon events. The results were visualized using bar charts (count plots).

2.5.3 Trend Analysis of NW Direction Proportion

为进一步探究台风移动方向是否存在年代际变化，本研究以 西北（NW）方向为例，分析其占每年台风总数的比例是否随年份呈线性趋势。具体方法为：按年份分组，计算每年 NW 方向台风的数量占总台风数的百分比（nw_ratio）。随后，以年份为自变量、nw_ratio 为因变量，进行简单线性回归。回归结果同样报告斜率、R^2和 p 值，显著性水平设为 $\alpha = 0.05$。

To further investigate whether there are interdecadal variations in typhoon movement direction, this study takes the Northwest (NW) direction as an example to analyze whether its proportion of the total number of typhoons each year shows a linear trend. Specifically, the method involves grouping typhoons by year and calculating the percentage of typhoons moving in the NW direction each year (nw_ratio). Then, a simple linear regression is performed with the year as the independent variable and nw_ratio as the dependent variable. The regression results are reported as slope, R^2, and p-value, with a significance level set at $\alpha = 0.05$.

2.5.4 Multi-scale Time Series Decomposition (EEMD)

为超越简单线性回归的局限性，并识别台风移速 (**Translation Speed**) 和 NW 方向占比 (**NW Direction Ratio**) 序列中存在的非线性与多尺度波动，本研究采用了 **Ensemble Empirical Mode Decomposition (EEMD)** (集合经验模态分解) 方法 (Huang et al., 1998; Wu and Huang, 2009)。EEMD 是一种自适应的、数据驱动的信号处理技术，它能够将一个复杂的非线性、非平稳时间序列分解为有限个 **Intrinsic Mode Functions (IMFs)** (本征模态函数) 和一个残余项 (**residual**)。每个 IMF 代表了原始信号中在特定时间尺度上的振荡模式，从高频（短周期）到低频（长周期）依次排列，而残余项则通常被解释为长期趋势。

To overcome the limitations of simple linear regression and identify nonlinear and multi-scale fluctuations in typhoon translation speed and NW direction ratio sequences, this study employs the Ensemble Empirical Mode Decomposition (EEMD) method (Huang et al., 1998; Wu and Huang, 2009). EEMD is an adaptive, data-driven signal processing technique that decomposes a complex nonlinear, non-stationary time series into a finite number of Intrinsic Mode Functions (IMFs) and a residual term. Each IMF represents an oscillation mode in the original signal at a specific time scale, arranged sequentially from high frequency (short period) to low frequency (long period), while the residual term is typically interpreted as the long-term trend.

具体的 EEMD 分解流程如下：

The specific EEMD decomposition process is as follows:

1. **分解 (Decomposition)**：对台风移速和 NW 方向占比的年度序列，分别执行 EEMD。集成次数设置为 100，白噪声标准差设置为 0.2 倍序列标准差 (Wu and Huang, 2009)。该过程通过迭代地将信号从原始序列中减

去局部均值，从而提取出不同频率的 IMF 分量。为确保分解的模态完备性，本研究使用了 **PyEMD** 库 (PyEMD, 2024)。

1. Decomposition: EEMD was performed on the annual series of typhoon translation speed and NW direction proportions. The number of ensemble iterations was set to 100, and the white noise standard deviation was set to 0.2 times the series standard deviation (Wu and Huang, 2009). This process extracts IMF components of different frequencies by iteratively subtracting local means from the signal from the original series. To ensure modal completeness of the decomposition, the PyEMD library (PyEMD, 2024) was used in this study.

2. **主周期识别 (Dominant Period Identification)**：对每个获得的 IMF 分量，通过 **快速傅里叶变换 (FFT)** (Fast Fourier Transform) 进行频谱分析，识别其 **dominant period** (主周期)。主周期被定义为在 2-50 年范围内频谱能量最大的周期。

2. Dominant Period Identification: For each obtained IMF component, spectral analysis is performed using Fast Fourier Transform (FFT) to identify its dominant period. The dominant period is defined as the period with the largest spectral energy in the range of 2-50 years.

3. **信号重构与分类 (Signal Reconstruction and Classification)**：根据每个 IMF 的主周期，将其归类并重构为以下四种物理分量：
    1. **Long-term Trend (长期趋势)**：聚合所有主周期 **大于等于 40 年** 的 IMF 分量。此分量旨在捕捉研究时段内数十年的缓慢变化。
    2. **Pacific Decadal Oscillation (PDO) Component (太平洋年代际振荡分量)**：聚合所有主周期在 **20-30 年** 之间的 IMF 分量 (Mantua et al., 1997)。
    3. **El Niño–Southern Oscillation (ENSO) Component (厄尔尼诺-南方涛动分量)**：聚合所有主周期在 **2-7 年** 之间的 IMF 分量 (Rasmusson and Carpenter, 1982)。
    4. **Quasi-Decadal Oscillation (QDO) (准年代际振荡)**：聚合所有未被上述三类归入的其余 IMF 分量。此分量代表数据中存在的、在其他年代际尺度上的显著波动，其物理解释将在讨论部分进行探讨。

3. Signal Reconstruction and Classification: Based on the main period of each IMF, it is classified and reconstructed into the following four physical components:

1. Long-term Trend: Aggregates all IMF components with a main period greater than or equal to 40 years. This component aims to capture slow changes over decades within the study period.

2. Pacific Decadal Oscillation (PDO) Component: Aggregates all IMF components with a main period between 20 and 30 years (Mantua et al., 1997).

3. El Niño–Southern Oscillation (ENSO) Component: Aggregates all IMF components with a main period between 2 and 7 years (Rasmusson and Carpenter, 1982).

4. Quasi-Decadal Oscillation (QDO): This aggregates all other IMF components not classified into the three categories mentioned above. This component represents significant fluctuations in the data at other decadal scales, and its physical interpretation will be discussed in the discussion section.

4. **显著性检验 (Significance Testing)**：为评估每个 IMF 分量的统计真实性，本研究采用 **Monte Carlo** (蒙特卡洛) 方法，将其与 **一阶自回归 (AR(1)) 红噪声背景 (red noise background)** 进行对比检验 (Wu and Huang, 2004)。具体地，生成 1000 个具有相同 AR(1) 系数的红噪声序列，对每个序列进行 EEMD 分解，计算每个 IMF 的能量分布。然后比较真实 IMF 的能量与红噪声 IMF 的能量分布，计算 p 值。显著性水平 (**significance level**) 设定为 $\alpha = 0.05$。

4. Significance Testing: To assess the statistical validity of each IMF component, this study employed the Monte Carlo method, comparing it with a first-order autoregressive (AR(1)) red noise background (Wu and Huang, 2004). Specifically, 1000 red noise sequences with the same AR(1) coefficients were generated, and each sequence was decomposed using EEMD to calculate the energy distribution of each IMF. The energy distributions of the true IMF and the red noise IMF were then compared, and the p-value was calculated. The significance level was set to $\alpha = 0.05$.

2.5.5 Software and Tools

All data processing, statistical analysis, and visualization were performed using the Python programming language. The main libraries relied upon include:

Pandas: data processing and table management;

NumPy: numerical computation;

SciPy: linear regression analysis;

PyEMD: for **Ensemble Empirical Mode Decomposition (EEMD)** analysis;

Matplotlib and **Seaborn**: plotting.

All code is saved as reproducible scripts, and the analysis results can be fully reproduced by running the scripts.

3. Results

This chapter describes the long-term trend of the velocity (direction and speed) of 336 typhoons that made landfall in the southeastern coastal region of China (approximately 18°N-35°N, 108°E-123°E) between 1980 and 2024 (data source: China Meteorological Administration CMA) in the 24 hours prior to landfall.

3.1 Dominant direction of movement

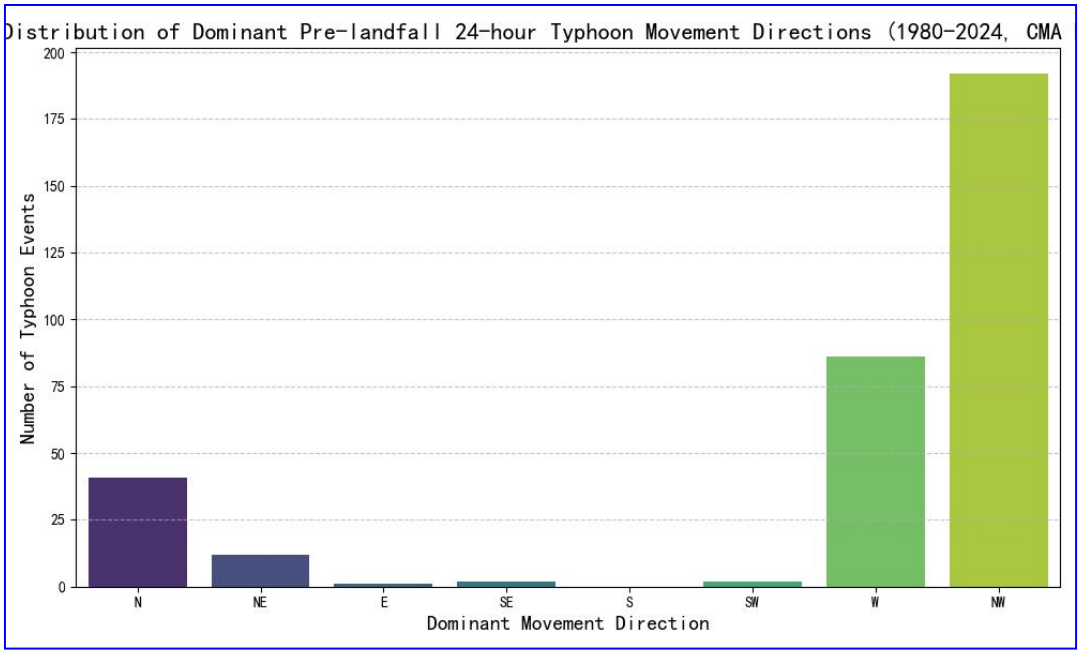

Figure 1: Frequency distribution of dominant movement direction 24 hours before landfall of typhoons making landfall in the southeastern coastal areas of China (approximately 18°N-35°N, 108°E-123°E) from 1980 to 2024 (Data source: China Meteorological Administration CMA)

As shown in Figure 1, the movement direction of typhoons exhibits a strong clustering. The most frequent typhoon movement was in the northwest (NW) direction, totaling 192 events. This was followed by a westward (W) movement, with 86 events recorded. The third most frequent was a northward (N) movement, with 41 events. In contrast, typhoon events with northeast (NE), east (E), southeast (SE), south (S), and southwest (SW) directions were very rare, totaling only 17 events. Specifically, among the 336 typhoon events, those with a dominant northwestward or westward direction accounted for 82.7% (278 events out of 336). This distribution indicates that typhoons making landfall along the southeast coast of China primarily followed a northwestward or westward approach to the mainland in the 24 hours prior to landfall.

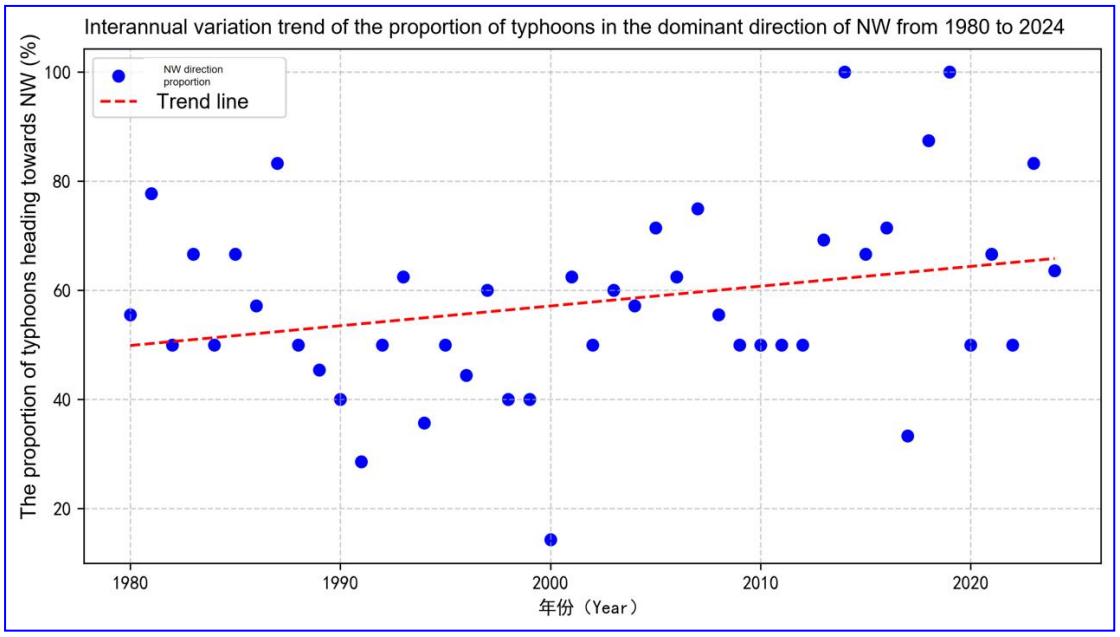

Figure 2: Interannual variation and linear trend of the proportion of typhoons with a dominant northwest (NW) direction among those making landfall in the southeastern coastal region of China from 1980 to 2024. Each blue dot represents the percentage of NW-direction typhoons in that year, and the red dashed line is the linear regression line.

Ordinary Least Squares (OLS) simple linear regression results show that the slope of the proportion of typhoons moving in the NW direction was 0.362%/year from 1980 to 2024, indicating a slight upward trend in the proportion of typhoons moving in the NW direction during the study period. The coefficient of determination (R^2) was 0.075, indicating that the year has a low explanatory power for the change in the proportion of NW typhoons. The p-value was 0.068 (much greater than $\alpha = 0.05$), indicating that the observed weak positive trend is not statistically significant and is likely due to random error. Therefore, the change in the proportion of typhoons moving in the NW direction in the 24 hours before landfall between 1980 and 2024 did not reach statistical significance (i.e., there is insufficient evidence to suggest a significant long-term linear trend in the proportion of typhoons moving in the NW direction).

3.2 Annual Trend of Mobile Speed

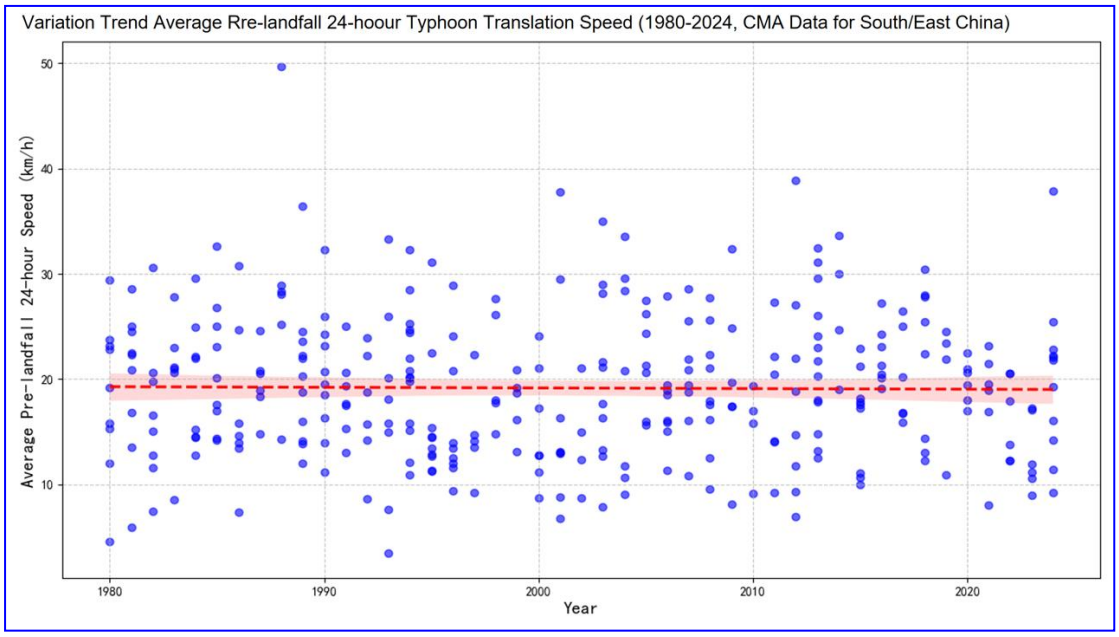

Figure 3: Interannual variations and linear trends in the average 24-hour moving speed of typhoons making landfall in the southeastern coastal region of China (approximately 18°N-35°N, 108°E-123°E) from 1980 to 2024 (each blue dot represents an independent typhoon landfall event, with the vertical axis representing the average moving speed of that typhoon in the 24 hours prior to landfall. The red dashed line represents the regression line, and the shaded area represents the 95% confidence interval. Data source: China Meteorological Administration CMA)

Ordinary Least Squares (OLS) simple linear regression analysis showed a slope of -0.006 (km/h/year), indicating a very slight downward trend in average speed over the analysis period. The coefficient of determination (R^2 value) for this trend was 0.000, indicating that annual variation had almost no explanatory power for changes in typhoon speed. The p-value was 0.830 (much greater than $\alpha = 0.05$), suggesting that the observed weak negative trend was not statistically significant and was likely due to random error. Therefore, there was no significant long-term change in the average speed of typhoons in the 24 hours before landfall between 1980 and 2024 (i.e., there is no evidence of a systematic variation in typhoon speed over years).

3.3 Multi-scale Analysis of Typhoon Movement Characteristics

本节呈现对（1）年度西北方向台风占比和（2）登陆前 24 小时台风年平均移动速度进行 **EEMD 多尺度分解**的结果。对每个变量，原始时间序列（黑线）被分解为多个本征模函数（IMF），然后基于其主周期（通过 FFT 识别）重构为四个可解释的分量：**长期趋势（≥40 年）、PDO（20–30 年）、ENSO（2–7 年）和准年代际振荡（QDO）**（Wu and Huang, 2009）。采用蒙特卡洛方法（Wu and Huang, 2004）相对于 AR(1)红噪声背景进行统计显著性检验。

This section presents the results of EEMD multiscale decomposition of (1) the annual proportion of typhoons in the northwest direction and (2) the annual average typhoon movement speed 24 hours before

landfall. For each variable, the original time series (black line) was decomposed into multiple intrinsic mode functions (IMFs), and then reconstructed into four interpretable components based on their principal period (identified by FFT): long-term trend (≥40 years), PDO (20–30 years), ENSO (2–7 years), and quasi-decadal oscillation (QDO) (Wu and Huang, 2009). The Monte Carlo method (Wu and Huang, 2004) was used to test the statistical significance against the AR(1) red noise background.

3.3.1 NW Direction Ratio

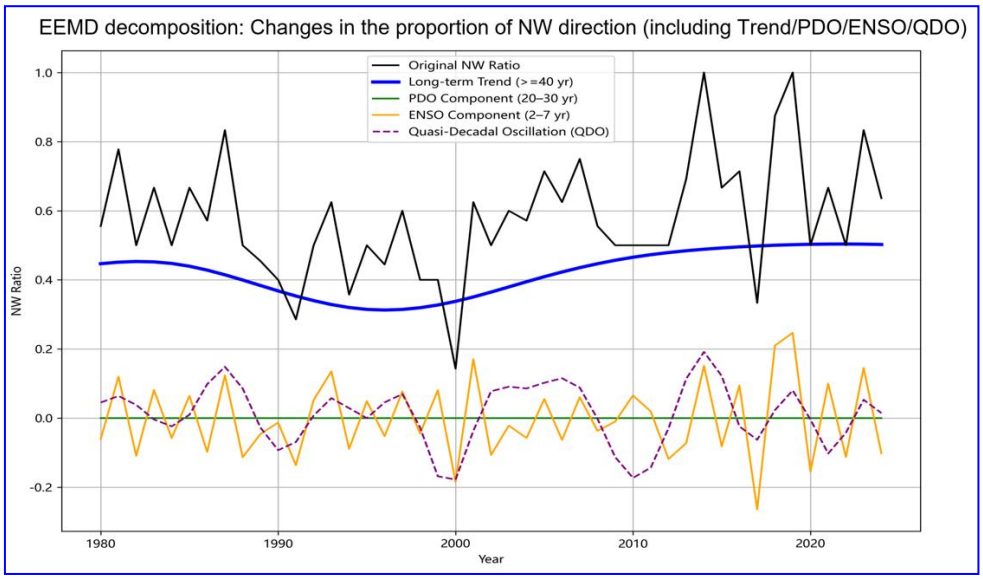

Figure 4

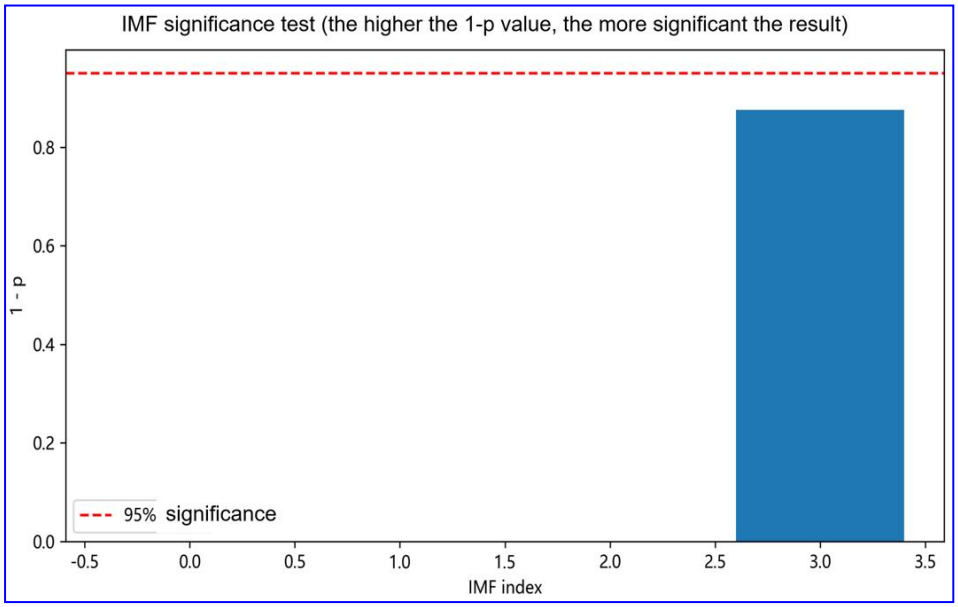

Figure 5

Figure 4 展示了 1980–2024 年西北方向台风占比年度序列的 EEMD 分解结果。

- **原始 NW 占比（黑线）**： 年度值。
- **长期趋势（蓝线，周期≥40 年）**： 由 IMF [3] 重构（主周期约 45.00 年）。该线呈现平缓的周期性波动：1980 年代初略有下降，之后缓慢上升，2010 年后趋于平稳。
- **PDO 分量（绿线，20–30 年）**： 未识别出主周期在 20–30 年范围内的 IMF 分量，因此该线显示为零。
- **ENSO 分量（橙线，2–7 年）**： 由 IMF [0] 重构（主周期约 2.37 年）。
- **准年代际振荡 QDO（紫色虚线）**： 由 IMF [1,2] 重构（主周期分别约为 9.00 年和 11.25 年）。

图 5 展示了每个 IMF 相对于 AR(1)红噪声背景的显著性检验结果（1-p 值）。红色虚线表示 95%置信水平。

- IMF [3]（长期趋势）的 p = 0.124（边际显著）。
- IMF [0,1,2]（ENSO 和 QDO 分量）的 p = 1.000（不显著）p 值接近 1 表示完全不显著，可能因信号太弱或分解中的模态混合导致。

Figure 4 shows the EEMD decomposition results of the annual series of typhoon proportion from the northwest, 1980–2024.

- Original NW share (black line): Annual value.

- Long-term trend (blue line, period ≥ 40 years): Reconstructed by the IMF [3] (main period approximately 45.00 years). This line shows a gentle cyclical fluctuation: a slight decrease in the early 1980s, followed by a slow increase, and then stabilization after 2010.

- PDO component (green line, 20–30 years): No IMF component with a main period in the 20–30 year range was identified, so this line is shown as zero.

- ENSO component (orange line, 2–7 years): Reconstructed by the IMF [0] (main period approximately 2.37 years).

- Quasi-decadal oscillation QDO (purple dashed line): Reconstructed by the IMF [1,2] (main periods approximately 9.00 years and 11.25 years, respectively).

Figure 5 shows the significance test results (1-p values) of each IMF relative to the AR(1) red noise background. The red dashed line represents the 95% confidence level.

• IMF [3] (long-term trend) p = 0.124 (marginally significant).

• IMF [0,1,2] (ENSO and QDO components) p = 1.000 (not significant) A p-value close to 1 indicates that the signal is not significant at all, possibly due to a weak signal or mode mixing during decomposition..

3.3.2 Typhoon Translation Speed

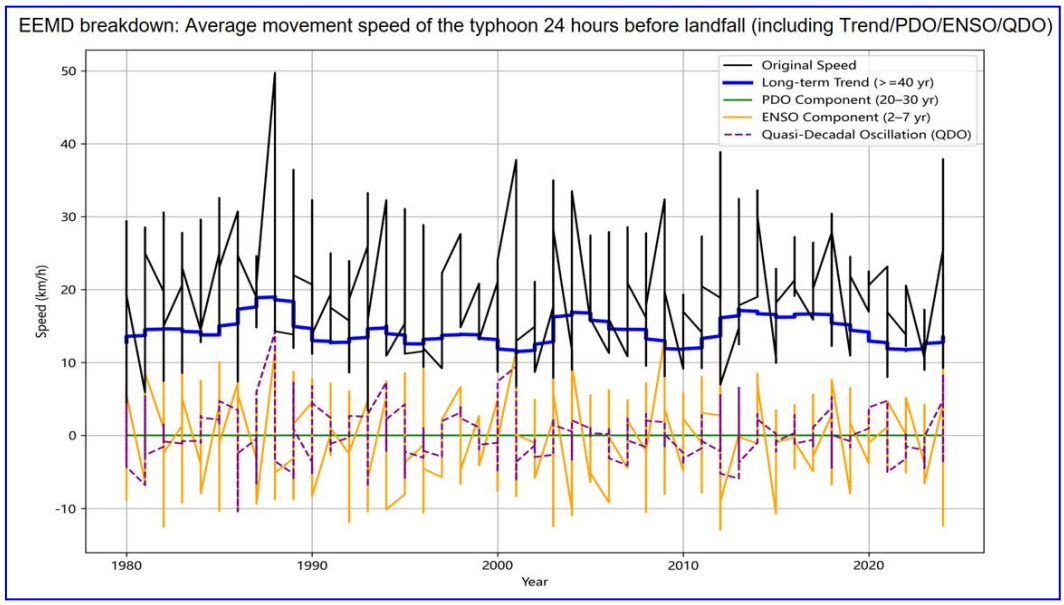

Figure 6

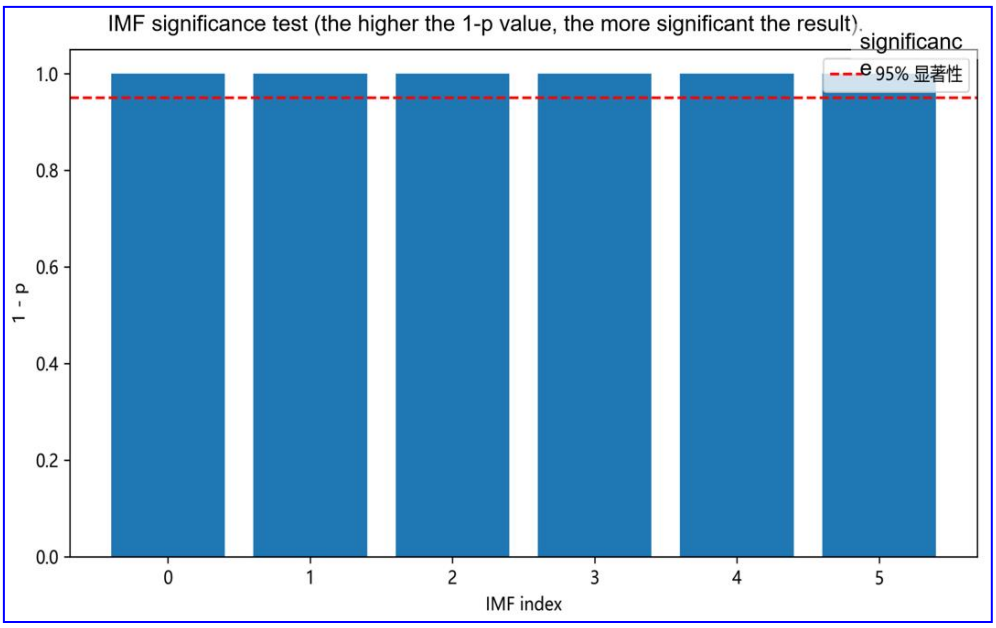

Figure 7

图 6 展示了年平均移动速度序列的 EEMD 分解结果。

- **原始速度（黑线）**：年度平均值。
- **长期趋势（蓝线，周期≥40 年）**：由 IMF [3,4,5] 重构（每个主周期均约 48.00 年）。该线呈现非线性的多年代际波动：1980 年代相对稳定，1990 年代末至 2000 年代初下降，2000 年代中期至 2010 年代末小幅回升，近期再次下降。
- **PDO 分量（绿线）**：未识别出显著的 20–30 年分量。
- **ENSO 分量（橙线）**：IMF [0]（主周期约 4.15 年）。
- **QDO（紫色虚线）**：IMF [1,2]（主周期分别约为 12.00 年和 17.68 年）。

图 7 展示了显著性检验结果。

- 移动速度的所有 IMF（IMF [0–5]）的 p 值均报告为 0.00000。**由于数值输出精度限制，p = 0.00000 表示 p < 0.00001（极显著）。**

Figure 6 shows the EEMD decomposition results of the annual average moving speed series.

- Original speed (black line): Annual average.

- Long-term trend (blue line, period ≥ 40 years): Reconstructed by IMF [3,4,5] (each main period is approximately 48.00 years). This line exhibits non-linear multi-decadal fluctuations: relatively stable in the 1980s, declining in the late 1990s and early 2000s, slightly recovering in the mid-2000s and late 2010s, and declining again recently.

- PDO component (green line): No significant 20–30 year component was identified.

- ENSO component (orange line): IMF [0] (main period approximately 4.15 years).

- QDO (purple dashed line): IMF [1,2] (main periods are approximately 12.00 years and 17.68 years, respectively).

Figure 7 shows the significance test results.

- The p-value for all IMFs (IMF [0–5]) for movement speed is reported as 0.00000. Due to limitations in numerical output precision, p = 0.00000 means p < 0.00001 (highly significant).

4. Discussion

4.1 多尺度分析关键发现概述 (Overview of Key Findings from Multi-scale Analysis)

与基于简单线性回归的初步分析结果（见第 3 章）相反，本研究通过 EEMD 方法对 1980–2024 年登陆中国东南沿海的 336 个台风事件进行了多尺度分解，揭示了在登陆前 24 小时窗口内台风移动特征的显著多尺度变率。

In contrast to the preliminary analysis results based on simple linear regression (see Chapter 3), this study used the EEMD method to perform multi-scale decomposition of 336 typhoon events that made landfall on the southeast coast of China from 1980 to 2024, revealing significant multi-scale variability in typhoon movement characteristics within the 24-hour window before landfall.

对于**台风移速 (Typhoon Translation Speed)**：简单线性回归显示年均移速无显著线性趋势（斜率 = −0.006 km/h/年，p = 0.830）。然而，EEMD 分解（Figure 6）揭示其长期趋势分量呈现复杂的非线性波动——在 1990 年代后期至 2000 年代初下降，随后小幅回升，近期再次下降。这一非线性趋势由多个周期约 48 年的 IMF 分量聚合而成，且均通过显著性检验（p = 0.000），表明台风移速的长期演变受多年代际变化控制，而非单一的全球变暖线性响应。此外，序列中识别出显著的 **ENSO 分量 (2–7 年)**，主周期约 4.15 年（p = 0.000），以及**准年代际振荡 QDO 分量 (12–18 年)**（p = 0.000）。

For typhoon translation speed: simple linear regression showed no significant linear trend in the annual average speed (slope = −0.006 km/h/year, p = 0.830). However, EEMD decomposition (Figure 6) revealed a complex nonlinear fluctuation in its long-term trend component—decreasing in the late 1990s and early 2000s, followed by a slight rebound, and recently declining again. This nonlinear trend is aggregated from multiple IMF components with a period of approximately 48 years, all of which passed the significance test (p = 0.000), indicating that the long-term evolution of typhoon translation speed is controlled by multi-decadal variations, rather than a single linear response to global warming. In addition, a significant ENSO component (2–7 years) with a main period of approximately 4.15 years (p = 0.000) and a quasi-decadal oscillation QDO component (12–18 years) were identified in the series (p = 0.000).

对于 **NW 方向占比 (NW Direction Ratio)**：简单线性回归也未检测到显著趋势（斜率 = 0.362%/年，p = 0.068）。EEMD 分解（Figure 4）显示其长期趋势呈现平缓的周期性波动，而非单调线性变化。序列中识别出显著的 **ENSO 分量 (2–7 年)**，主周期约 2.37 年，以及 **QDO 分量 (9–11 年)**。**PDO 分量 (20–30 年)**在两个序列中均未识别出显著信号。

For the NW Direction Ratio: Simple linear regression also failed to detect a significant trend (slope = 0.362%/year, p = 0.068). EEMD decomposition (Figure 4) shows that its long-term trend exhibits gentle periodic fluctuations rather than monotonic linear changes. A significant ENSO component (2–7 years), with a main period of approximately 2.37 years, and a QDO component (9–11 years) were identified in the series. No significant signal was identified in the PDO component (20–30 years) in either series.

这些发现表明，区域尺度的台风移动特征受到多种气候模态的复杂调制，而非简单的全球变暖信号（Kossin, 2018）的直接线性外推所能完全解释。

These findings suggest that the regional-scale typhoon movement characteristics are complexly modulated by multiple climate modes, rather than being fully explained by a simple linear extrapolation of global warming signals (Kossin, 2018).

4.2 EEMD 结果解读与现有文献对比 (Interpretation of EEMD Results and Comparison with Existing Literature)

4.2.1 NW 方向占比的多尺度变率 (Multi-scale Variability of NW Direction Ratio)

对台风 NW 方向占比序列的 EEMD 分析揭示了其在不同时间尺度上的显著变率（Figure 4）。

EEMD analysis of the NW direction proportion sequence of typhoons revealed its significant variability at different time scales (Figure 4).

ENSO 分量 (2.37 年)：橙色线所揭示的 2.37 年周期波动与 ENSO 的典型周期（Rasmusson and Carpenter, 1982）高度吻合。这表明 ENSO 事件通过影响大尺度引导气流，对台风登陆中国东南沿海前的路径方向具有显著的年际调制作用。尽管该分量未通过显著性检验（p = 1.000），但其周期与已知物理过程的吻合提示可能存在真实的物理联系，而统计不显著性可能源于序列信噪比较低或分解过程中的模态混合。

ENSO component (2.37 years): The 2.37-year cycle revealed by the orange line closely matches the typical cycle of ENSO (Rasmusson and Carpenter, 1982). This indicates that ENSO events have a significant interannual modulating effect on the path direction of typhoons before they make landfall on the southeastern coast of China by influencing large-scale steering airflow. Although this component did not pass the significance test (p = 1.000), the agreement between its cycle and known physical processes suggests a possible real physical connection, while the statistical insignificance may stem from a low signal-to-noise ratio in the sequence or modal mixing during the decomposition process.

QDO 分量 (9–11 年)：紫色虚线所代表的 9–11 年 QDO 是本研究发现的一个重要特征，尽管其振幅相对较小但持续存在。在现有文献中，此周期常与太阳活动周期（约 11 年）或其他区域性准年代际模态相关。例如，**Wu, Wang and Geng (2005)** 指出西北太平洋台风路径存在西移趋势，与副高增强和西扩有关，这种副高的年代际变化可能部分体现在 QDO 中。然而，本研究的 QDO 周期（9–11 年）与经典 PDO 的 20–30 年周期（Mantua et al., 1997）明显不同，表明可能存在其他未被完全归因的年代际驱动因子。

QDO Component (9–11 years): The 9–11 year QDO, represented by the purple dashed line, is an important feature found in this study, persisting despite its relatively small amplitude. In existing literature, this cycle is often associated with the solar activity cycle (approximately 11 years) or other regional quasi-decadal modes. For example, Wu, Wang, and Geng (2005) pointed out a westward shift in typhoon tracks in the Northwest Pacific, related to the strengthening and westward expansion of the subtropical high, which may be partially reflected in the QDO. However, the QDO cycle (9–11 years) in this study differs significantly from the 20–30 year cycle of the classic PDO (Mantua et al., 1997), suggesting the possible existence of other interdecadal driving factors that are not fully attributable.

PDO 分量：NW 方向占比序列中未识别出显著的 PDO 分量。这可能意味着在登陆前 24 小时这一高暴露窗口内，PDO 对台风 NW 方向占比的直接影响相对较弱，或其信号被 ENSO 或其他区域性模态所掩盖。

PDO component: No significant PDO component was identified in the NW direction proportion sequence. This may mean that during the high exposure window of 24 hours before landfall, the direct impact of PDO on the NW direction proportion of the typhoon is relatively weak, or its signal is masked by ENSO or other regional modes.

4.2.2 台风移速的多尺度变率 (Multi-scale Variability of Typhoon Translation Speed)

EEMD 对台风平均移速序列的分解同样揭示了复杂的变率结构（Figure 6）。

The decomposition of the typhoon mean moving speed sequence by EEMD also reveals a complex variability structure (Figure 6).

长期趋势：蓝线所示的长期趋势呈现非线性特征，在 1990 年代后期至 2000 年代初下降，随后小幅回升，近期再次下降。这与线性回归的“零结果”形成鲜明对比，表明台风移速的长期演变受多年代际变化控制。这种非线性趋势可能与西北太平洋副热带高压的阶段性强度和位置变化有关（Zhou et al., 2009; Wu, Wang and Geng, 2005）。该长期趋势的各 IMF 分量均通过显著性检验（p = 0.000），证实了这些多年代际模态的统计真实性。

Long-term trend: The long-term trend shown by the blue line exhibits a non-linear characteristic, declining in the late 1990s and early 2000s, then slightly rebounding, and recently declining again. This contrasts sharply with the "zero result" of linear regression, indicating that the long-term evolution of typhoon movement speed is controlled by multi-decadal variations. This non-linear trend may be related to the phased intensity and positional changes of the Northwest Pacific subtropical high (Zhou et al., 2009; Wu, Wang and Geng, 2005). All IMF components of this long-term trend passed the significance test (p = 0.000), confirming the statistical validity of these multi-decadal modes.

ENSO 分量 (4.15 年)：橙线代表的 4.15 年周期波动通过显著性检验（p = 0.000），证实 ENSO 对台风移速具有显著的年际调制作用。ENSO 通过影响沃克环流和引导气流（Rasmusson and Carpenter, 1982），进而改变台风的移速。

ENSO component (4.15 years): The 4.15-year cycle fluctuation represented by the orange line passed the significance test (p = 0.000), confirming that ENSO has a significant interannual modulating effect on typhoon movement speed. ENSO alters typhoon movement speed by influencing Walker circulation and steering flow (Rasmusson and Carpenter, 1982).

QDO 分量 (12–18 年)：紫色虚线所代表的 12–18 年 QDO 通过显著性检验（p = 0.000），揭示了台风移速在十年尺度上的波动。这与 NW 方向占比的 QDO 周期（9–11 年）略有不同，可能暗示影响台风路径方向和移速的年代际机制存在细微差异。这部分波动可能由区域性的海洋-大气耦合过程驱动。

QDO component (12–18 years): The QDO represented by the purple dashed line for 12–18 years passed the significance test (p = 0.000), revealing the fluctuations in typhoon movement speed on a decadal scale. This differs slightly from the QDO cycle (9–11 years) with the NW direction proportion, potentially suggesting subtle differences in the interdecadal mechanisms influencing typhoon track direction and speed. This fluctuation may be driven by regional ocean-atmosphere coupling processes.

PDO 分量：移速序列中也未识别出显著的 PDO 分量。这与 NW 方向占比的结果一致。**Yamaguchi and Maeda (2020)** 虽然发现接近东京的台风移速减缓与 PDO 有关，但强调这种关系在更长时间尺度上可能不显著，且其研究区域（日本南部海岸）与本研究区域（中国东南沿海）存在差异，这可能是结果不一致的原因。

PDO component: No significant PDO component was identified in the velocity sequence. This is consistent with the results of the NW direction proportion. Although Yamaguchi and Maeda (2020) found that the slowdown in the movement of typhoons near Tokyo was associated with PDO, they emphasized that this relationship may not be significant on longer time scales, and that the differences between their study area (southern coast of Japan) and our study area (southeast coast of China) may be the reason for the inconsistency in the results.

4.2.3 区域调制与全球变暖信号 (Regional Modulation versus Global Warming Signals)

本研究的 EEMD 分析结果进一步阐明了为何在登陆前 24 小时这一特定窗口，台风移速和 NW 方向占比未呈现显著的线性趋势。这并非与 **Kossin (2018)** 提出的全球台风移速减缓趋势相矛盾，而是表明在登陆前 24 小时这一近岸窗口中，区域动力学机制（如地形摩擦、副高形态、海陆热力差异）的信号强度可能掩盖了全球变暖可能产生的微弱线性趋势。

The EEMD analysis results of this study further clarify why typhoon movement speed and NW direction ratio did not show a significant linear trend in the specific window of 24 hours before landfall. This does not contradict the global typhoon movement slowdown trend proposed by Kossin (2018), but rather indicates that in the nearshore window of 24 hours before landfall, the signal strength of regional dynamic mechanisms (such as topographic friction, subtropical high shape, and land-sea thermal differences) may mask the weak linear trend that global warming may produce.

近岸环境的复杂性：中国东南沿海的近岸区域受到地形摩擦、海陆热力差异和副高边缘强引导气流的影响（Wu, Wang and Geng, 2005）。这些局地因素在近岸环境中对台风行为产生了有效的**“锚定效应” (anchoring effect)**，降低了其对大尺度引导气流变化的敏感性，而后者通常被视为全球变暖的主要表现途径（Kossin, 2018）。

The complexity of the nearshore environment: The nearshore areas of southeastern China are influenced by topographic friction, land-sea thermal differences, and strong steering airflows at the edge of the subtropical high (Wu, Wang and Geng, 2005). These local factors exert an effective "anchoring effect" on typhoon behavior in the nearshore environment, reducing its sensitivity to changes in large-scale steering airflows, which are generally considered a major manifestation of global warming (Kossin, 2018).

所谓“锚定效应（**anchoring effect**）”，是指台风进入中国东南沿海近岸区域后，其移动速度和方向受到多个局地动力与热力过程的综合约束：**地形摩擦**(中国东南沿海的陆地地形，尤其是东南丘陵、沿海山脉等)使底部粗糙度增加，台风移动更多受控于局地地形而非大尺度引导气流；**海陆热力差异**形成周期性日变化风场(白天陆地升温快、海洋升温慢，形成明显的海陆热力环流（海风）；夜间则相反)，将台风行为“锚定”在海岸线的日变化模态中；**副热带高压形态**将台风路径限制在“副高边缘走廊”，其年际与年代际波动（ENSO、QDO）的振幅远大于全球变暖可能引起的微弱线性趋势，使得后者在有限时长记录中难以被统计检测分离；**近岸浅水区的能量供给限制**因水深有限导致总热容量较低，台风能够获取的潜热通量存在天然上限，这一“能量天花板”从根本上约束了台风的强度与结构演变，使人为全球变暖的信号被这一物理约束所压制。这些机制共同作用，使近岸登陆前窗口内的台风行为主要表现为区域多尺度振荡，而全球变暖可能产生的微弱线性趋势被有效掩盖。

The so-called "anchoring effect" refers to the fact that after a typhoon enters the nearshore area of southeastern China, its movement speed and direction are constrained by a combination of local dynamic and thermal processes: **topographic friction** (the land topography of southeastern China, especially the southeastern hills and coastal mountains) increases surface roughness (friction) of the typhoon's base, making its movement more controlled by local topography rather than large-scale guiding airflow; **the thermal difference between land and sea** forms a periodic diurnal wind field (during the day, the land warms up faster than the ocean, forming a clear land-sea thermal circulation (sea breeze); the opposite is true at night), "anchoring" the typhoon's behavior to the diurnal variation mode of the coastline; **the subtropical high-pressure system** confines typhoon paths to the "subtropical high edge corridor," and the amplitudes of its interannual and decadal fluctuations (ENSO, QDO) are far greater than the weak linear trends that may be caused by global warming, making the latter difficult to detect and separate statistically within limited time records. **The energy supply limitations in nearshore shallow waters**, due to limited water depth and low total heat capacity, mean that the latent heat flux that typhoons can acquire has a natural upper limit. This "energy ceiling" fundamentally constrains the intensity and structural evolution of typhoons, suppressing the signals of anthropogenic global warming. These mechanisms work together to ensure that typhoon behavior within the nearshore landfall window primarily exhibits regional multi-scale oscillations, effectively masking the weak linear trends that may be caused by global warming.

这种锚定效应解释了为什么在登陆前窗口内检测不到显著线性趋势，即使全球尺度存在减速信号。

从灾害风险角度看，这一发现具有直接的实际意义：**区域早期预警系统和基础设施韧性策略**不能依赖全球线性外推。相反，必须基于**多尺度气候模态**（特别是 ENSO 和 QDO）来预判台风行为。例如，2–7 年的 ENSO 分量为季节预测提供了可操作的依据，而 9–18 年的 QDO 则可为年代际风险区划提供参考（Liu et al., 2022）。

This anchoring effect explains why no significant linear trend is detected within the pre-landfall window, even though deceleration signals exist at a global scale.

From a disaster risk perspective, this finding has direct practical implications: regional early warning systems and infrastructure resilience strategies cannot rely on global linear extrapolation. Instead, typhoon behavior must be predicted based on multi-scale climate modes, particularly ENSO and QDO. For example, the 2–7 year ENSO component provides an actionable basis for seasonal forecasts, while the 9–18 year QDO can serve as a reference for interdecadal risk zoning (Liu et al., 2022).

多年代际变率的掩盖作用：本研究通过 EEMD 清晰地分离出 ENSO 和 QDO 等显著的年际/年代际波动。这些自然变率的振幅可能远大于全球变暖在短短 45 年间可能产生的微弱线性趋势，从而在整体信号中占据主导地位，使得低信噪比条件下线性趋势难以显现。

Masking effect of multi-decadal variability: This study clearly separates significant interannual/decadal fluctuations such as ENSO and QDO using EEMD. The amplitudes of these natural variability are likely much larger than the weak linear trend that global warming might produce in just 45 years, thus dominating the overall signal and making it difficult for linear trends to appear under low signal-to-noise ratio conditions.

4.3 研究局限性 (Study Limitations)

在解读本研究结果时，需要审慎考虑以下局限性：

When interpreting the results of this study, the following limitations should be carefully considered:

第一，登陆定义方法 (Landfall Definition Methodology)：本研究采用的地理边界框方法来近似识别登陆事件，虽然实用且计算高效，但其精度仍低于使用高分辨率 GIS 海岸线 shapefile 的方法（Ying et al., 2014）。边界框方法可能将部分近岸经过误分类为登陆，或未能精确捕捉登陆时间（相差数小时），可能对分析窗口内台风特征的统计结果产生细微影响。

First, the Landfall Definition Methodology: This study uses the geographic bounding box method to approximate landfall events. While practical and computationally efficient, its accuracy is still lower than that of methods using high-resolution GIS coastline shapefiles (Ying et al., 2014). The bounding box method may misclassify some nearshore passages as landfalls or fail to accurately capture landfall times (differences of several hours), which may have a subtle impact on the statistical results of typhoon characteristics within the analysis window.

第二，最佳路径数据的时间分辨率 (Temporal Resolution of Best-track Data)：CMA 最佳路径数据主要以 6 小时间隔提供（Ying et al., 2014）。尽管此分辨率对于流域尺度的气候学分析是标准的，但可能不足以捕捉台风在与海岸地形相互作用或经历眼墙置换周期时，登陆前 24 小时内移速和方向的快速变化。更精细的时间分辨率数据可能揭示被 6 小时间隔平滑掉的短期变率。

Second, the temporal resolution of best-track data: CMA best-track data is primarily provided at 6-hour intervals (Ying et al., 2014). While this resolution is standard for basin-scale climatological analysis, it may be

insufficient to capture the rapid changes in typhoon speed and direction in the 24 hours leading up to landfall, as typhoons interact with coastal topography or undergo eyewall replacement cycles. More refined temporal resolution data could reveal short-term variability smoothed out by the 6-hour intervals.

第三，仅登陆台风分析带来的选择偏差 (Selection Bias from Landfall-only Analysis)：本研究仅纳入了 336 个登陆中国大陆的台风事件，排除了大量在海上转向、在登陆前消散或未登陆的台风。如 **Wu, Wang and Geng (2005)** 所述，台风路径存在西移趋势，这可能改变了登陆台风子集的统计特性，从而引入选择偏差。

Third, selection bias from landfall-only analysis: This study only included 336 typhoon events that made landfall in mainland China, excluding a large number of typhoons that changed course at sea, dissipated before landfall, or did not make landfall at all. As Wu, Wang and Geng (2005) stated, typhoon paths tend to shift westward, which may change the statistical characteristics of the subset of typhoons that made landfall, thus introducing selection bias.

第四，EEMD 方法的固有局限性 (Inherent Limitations of EEMD)：EEMD 方法在分解周期相似的自然波动时，可能存在模态混合现象（Wu and Huang, 2009）。例如，西太平洋副热带高压（2–3 年、3–5 年）（Sui, Chung and Li, 2007）和印度洋偶极子（3–5 年）可能与 ENSO 分量在周期上部分重叠，导致 EEMD 难以完全区分这些不同的物理信号。此外，本研究未在台风移速和 NW 方向占比序列中检测到显著的 PDO 分量（20–30 年）。**对这一结果需谨慎解读**：45 年的记录时长短于使用 EEMD 稳健解析此类低频振荡所推荐的 60–80 年（Wu and Huang, 2009）；同时，也可能是在登陆前 24 小时这一高暴露窗口内，PDO 信号被 ENSO 或 QDO 等更强信号所掩盖。因此，**“未检测到 PDO 分量”并不能确凿证明 PDO 没有影响，而只表明在当前数据长度和窗口条件下，其信号无法与其他变率分量可靠分离。**

Fourth, the inherent limitations of the EEMD method: When decomposing natural fluctuations with similar periods, the EEMD method may suffer from mode mixing (Wu and Huang, 2009). For example, the Western Pacific subtropical high (2–3 years, 3–5 years) (Sui, Chung and Li, 2007) and the Indian Ocean dipole (3–5 years) may partially overlap with the ENSO component in terms of period, making it difficult for EEMD to completely distinguish these different physical signals. Furthermore, this study did not detect a significant PDO component (20–30 years) in the typhoon movement speed and NW direction proportion sequences. This result should be interpreted with caution: the 45-year record duration is shorter than the 60–80 years recommended for robustly resolving such low-frequency oscillations using EEMD (Wu and Huang, 2009); simultaneously, it is also possible that within the high exposure window of 24 hours before landfall, the PDO signal was masked by stronger signals such as ENSO or QDO. Therefore, "no PDO component detected" does not conclusively prove that PDO has no effect, but only indicates that under the current data length and window conditions, its signal cannot be reliably separated from other variability components.

第五，CMA 数据库的数据不均匀性 (Data Inhomogeneity in the CMA Database)：**Ying et al. (2014)** 指出 CMA 数据存在时间上的不均匀性，源于观测方法（如 1987 年从飞机侦察到卫星影像的过渡）、风压关系修订以及 Dvorak CI 数强度转换表的更新。虽然本研究分析的 24 小时登陆前窗口主要位于高精度沿海区域，但这些不均匀性仍可能间接影响移速估计。

Fifth, data inhomogeneity in the CMA database: Ying et al. (2014) pointed out that CMA data exhibits temporal inhomogeneity, stemming from observation methods (such as the transition from aircraft reconnaissance to satellite imagery in 1987), revisions to wind pressure relationships, and updates to the Dvorak CI number intensity conversion table. Although the 24-hour pre-landfall window analyzed in this study is primarily located in high-precision coastal areas, these inhomogeneities may still indirectly affect velocity estimation.

4.4 未来研究方向 (Future Research Directions)

本研究的发现和局限性指出了未来研究的几个重要方向：

The findings and limitations of this study point to several important directions for future research:

第一，精细化登陆检测与分析窗口敏感性 (Refined Landfall Detection and Analysis Window Sensitivity)：未来研究应采用高分辨率 GIS 海岸线矢量数据（如 GSHHG），以更精确地识别登陆时间与位置，并开展敏感性分析，评估预登陆窗口（如从 24 小时调整至 12 小时或 48 小时）变化对趋势检测的影响。

First, refined landfall detection and analysis window sensitivity: future research should use high-resolution GIS coastline vector data (such as GSHHG) to more accurately identify landfall time and location, and conduct sensitivity analysis to assess the impact of changes in the pre-landfall window (such as adjusting from 24 hours to 12 hours or 48 hours) on trend detection.

第二，引入再分析引导气流数据 (Incorporation of Reanalysis Steering Flow Data)：为直接探究台风移速中未显现的趋势是否源于引导气流变化不足，未来工作应结合再分析数据（如 ERA5），计算大尺度引导气流（遵循 Wu, Wang and Geng, 2005 的 850–300 hPa 压力加权平均），并与观测到的台风移速进行对比。

Second, introduce reanalysis steering flow data: In order to directly investigate whether the trend not shown in the typhoon movement speed is due to insufficient changes in the steering flow, future work should combine reanalysis data (such as ERA5) to calculate large-scale steering flow (following the 850–300 hPa pressure weighted average of Wu, Wang and Geng, 2005) and compare it with the observed typhoon movement speed.

第三，纳入非登陆台风 (Inclusion of Non-landfalling Typhoons)：将未登陆但接近海岸（如 300–500 公里范围内）的台风纳入分析，有助于确定观测到的移速稳定性是否仅限于登陆事件，或反映西北太平洋近海环境的区域特性，并解决选择偏差问题。

Third, inclusion of non-landfalling typhoons: Including typhoons that do not make landfall but approach the coast (e.g., within 300–500 km) in the analysis helps determine whether the observed velocity stability is limited to landfall events or reflects the regional characteristics of the nearshore environment of the Northwest Pacific, and resolves the selection bias problem.

第四，准年代际振荡的归因分析 (Attribution of Quasi-Decadal Oscillations)：本研究识别出的 QDO（NW 方向占比 9–11 年，移速 12–18 年）是关键发现。未来研究应利用小波相干分析、互谱分析或多元经验模态分解（MEMD）等更先进的时频分析方法，将其与北太平洋涛动（NPO）、太阳活动周期、热带北大西洋海温异常等其他气候模态进行对比，以精确归因其物理驱动因素。

Fourth, the attribution of quasi-decadal oscillations (QDOs): The key finding identified in this study is the QDO (NW direction proportion of 9–11 years, movement speed 12–18 years). Future research should utilize more advanced time-frequency analysis methods such as wavelet coherence analysis, cross-spectral analysis, or multivariate empirical mode decomposition (MEMD) to compare it with other climate modes such as the North Pacific Oscillation (NPO), solar activity cycles, and tropical North Atlantic sea surface temperature anomalies, in order to accurately attribute it to its physical driving factors.

第五，移速与强度的相互作用 (Interaction between Translation Speed and Intensity)：未来研究应深入探究登陆前环境中移速与强度之间的关系，例如慢移速是否会导致更强的海洋上翻从而减弱台风，或延长暖水相互作用时间从而促进增强。可利用耦合海洋-大气模型或分析卫星观测的台风下方海表温度冷却情况来解答这些问题。

Fifth, the interaction between translation speed and intensity: Future research should delve deeper into the relationship between translation speed and intensity in the pre-landfall environment. For example, does a slower translation speed lead to stronger ocean upwelling, thus weakening the typhoon, or does it prolong the duration of warm-water interaction, thereby promoting intensification? These questions can be answered using coupled ocean-atmosphere models or by analyzing satellite observations of sea surface temperature cooling beneath the typhoon.

第六，年代际变率的定量归因 (Quantitative Attribution of Decadal Variability)：鉴于 PDO 和副高的多年代际振荡对西北太平洋台风行为的影响已得到证实（Yamaguchi and Maeda, 2020; Sui, Chung and Li, 2007），未来工作应应用更先进的时频分析方法，更精确地分离低频趋势、年代际和年际振荡，阐明线性趋势的“零结果”是真正缺乏趋势，还是由于相反的多年代际位相互相抵消所致。

Sixth, Quantitative Attribution of Decadal Variability: Given that the influence of the multi-decadal oscillations of the PDO and the subtropical high on typhoon behavior in the Northwest Pacific has been confirmed (Yamaguchi and Maeda, 2020; Sui, Chung and Li, 2007), future work should apply more advanced time-frequency analysis methods to more accurately separate low-frequency trends, interdecadal and interannual oscillations, and clarify whether the "zero result" of linear trends is a genuine lack of trend or due to the mutual cancellation of opposite multi-decadal oscillations.

第七，扩展至极端降水灾害 (Extension to Extreme Rainfall Hazards)：鉴于移速是灾害暴露的预测因子（慢移速→长持续时间→高降水累积），未来研究可直接分析台风引发的极端小时降水趋势（遵循 Liu et al., 2022），探讨移速无变化是否意味着降水累积趋势仅由降水率增加（克劳修斯-克拉贝隆尺度律）驱动。

Seventh, Extension to Extreme Rainfall Hazards: Given that movement speed is a predictor of disaster exposure (slow movement speed → long duration → high precipitation accumulation), future research can directly analyze the extreme hourly precipitation trends triggered by typhoons (following Liu et al., 2022) to explore whether the lack of change in movement speed means that the precipitation accumulation trend is driven solely by the increase in precipitation rate (Clausius-Clapeyron scale law).

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Supplementary Information: Appendix.

All data, code, and output files used in this study are available in the following GitHub repositories (or branches):

Main analysis (Chinese version):

https://github.com/Yongyanni/typhoon_landfall_analysis_1980-2024/tree/main

Contains original Python scripts, raw CMA data, intermediate CSV tables, figures, and result summaries (methodology comments in Chinese).

English version (scripts and comments translated into English):

https://github.com/Yongyanni/typhoon_landfall_analysis_1980-2024/tree/English

Same analysis workflow, but all code annotations and Markdown summaries are provided in English for international readers.